Graphs And Transformations

Solutions

starter

Solution 2

1(a) Factorise $x^2 + 6x + 5$.

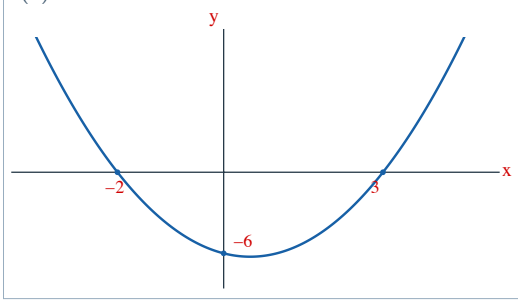
1(a) $(x + 5)(x + 1)$

1(b) Factorise $x^2 - 4x + 3$.

1(b) $(x - 3)(x - 1)$

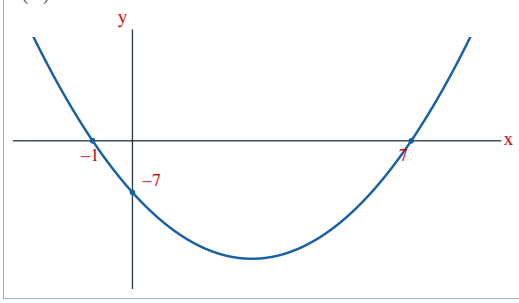
2(a) Sketch $y = (x + 2)(x - 3)$.

2(a)



2(b) Sketch $y = x^2 - 6x - 7$.

2(b)



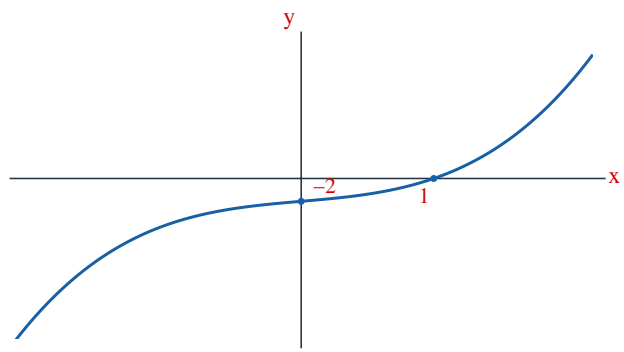
3 Complete a table and draw $y = x^3 + x - 2$ for $x = -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2$.

3

x	-2	-1.5	-1	$-.5$	0
y	-12	-6.875	-4	-2.625	-2

x	$.5$	1	1.5	2
y	-1.375	0	2.875	8

3 sketch



Chapter Overview

Solution 3

1. Sketching graphs

Cubic, quartic and reciprocal functions: roots, intercepts, repeated roots and end behaviour.

2. Points of intersection

Connect graphical intersections with simultaneous equations.

3. Transformations

Translate, stretch and reflect graphs. Track points, intercepts and asymptotes.

Polynomial Graphs

Solution 4

A polynomial has nonnegative integer powers of x:

$a + bx + cx^2 + dx^3 + ex^4 + \cdots$

Its **degree** is the highest power with a nonzero coefficient.

Degree 0

4

Degree 0

Constant

Degree 1

$2x - 1$

Degree 1

Linear

Degree 2

$x^2 + 3$

Degree 2

Quadratic

Degree 3

x^3

Degree 3

Cubic

Degree 4

x^4

Degree 4

Quartic

Degree 5

x^5

Degree 5

Quintic

Polynomial Graphs — Solution 5

Turning points How does the degree restrict the shape?

Turning points

A polynomial of degree n has **at most** n−1 turning points. It can have fewer: $y = x^4$ has just one.

End behaviour What determines the direction of the tails?

End behaviour

The leading term dominates for large $|x|$. For example, if $y = ax^2 + bx + c$ and $a > 0$, then $y \to \infty$ as $x \to \pm\infty$.

An upward quartic need not have three turns. Do not infer the number of turns just from its degree.

Polynomial Graphs — Solution 6

For $y = ax^n + \cdots$, with $a \neq 0$, describe the left and right tails.

Degree 2, a>0

$y = ax^2 + \cdots$

Degree 2, a>0 Both tails upwards.

Degree 2, a<0

$y = ax^2 + \cdots$

Degree 2, a<0 Both tails downwards.

Degree 3, a>0

$y = ax^3 + \cdots$

Degree 3, a>0 Left tail down; right tail up.

Degree 3, a<0

$y = ax^3 + \cdots$

Degree 3, a<0 Left tail up; right tail down.

Degree 4, a>0

$y = ax^4 + \cdots$

Degree 4, a>0 Both tails upwards.

Degree 4, a<0

$y = ax^4 + \cdots$

Degree 4, a<0 Both tails downwards.

Degree 5, a>0

$y = ax^5 + \cdots$

Degree 5, a>0 Left tail down; right tail up.

Degree 5, a<0

$y = ax^5 + \cdots$

Degree 5, a<0 Left tail up; right tail down.

Cubics — Solution 7

1 Sketch

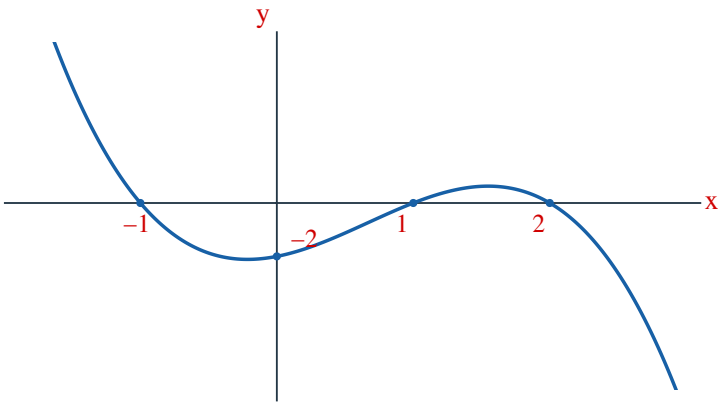
$y = (x-2)(1-x)(1+x)$

1 features

Leading term $-x^3$: left up, right down.

Crosses at $x = -1, 1, 2$; y-intercept $(0, -2)$.

1 sketch



2 Sketch

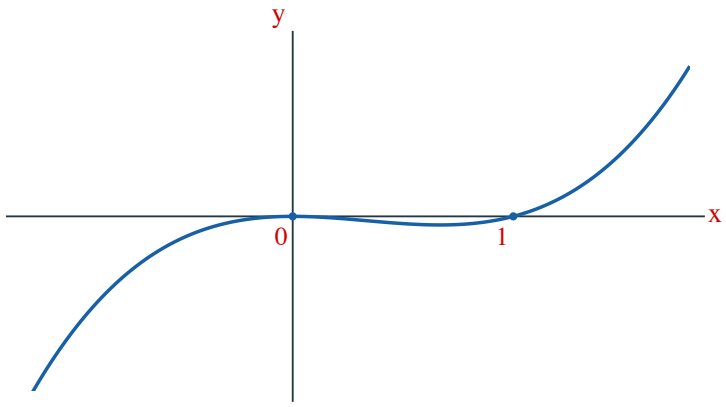
$y = x^2(x-1)$

2 features

Leading term x^3: left down, right up.

Touches at $(0, 0)$; crosses at $(1, 0)$.

2 sketch



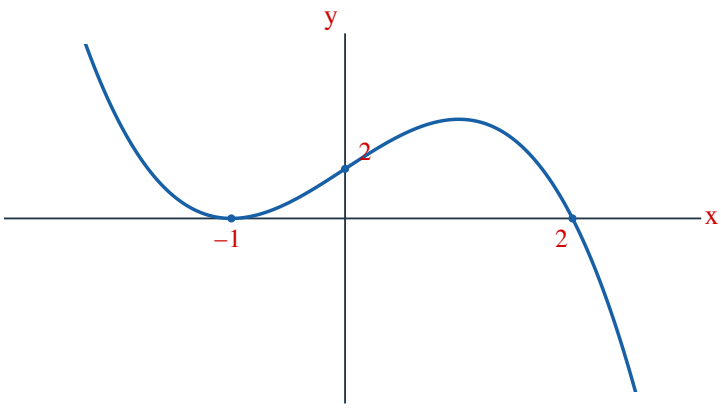
Cubics — Solution 8

1 Sketch

$y = (2-x)(x+1)^2$

1 features Leading term $-x^3$. Touches at $x = -1$, crosses at $x = 2$; y-intercept $(0, 2)$.

1 sketch



2 Sketch

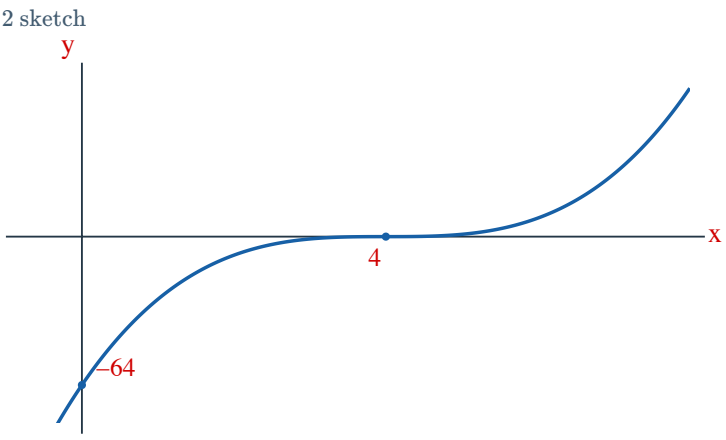
$y = (x-4)^3$

2 features

Leading term x^3. Triple root at 4: crosses with a horizontal tangent.

y-intercept $(0, -64)$.

2 sketch



Cubics with Limited Roots

Solution 9

Sketch

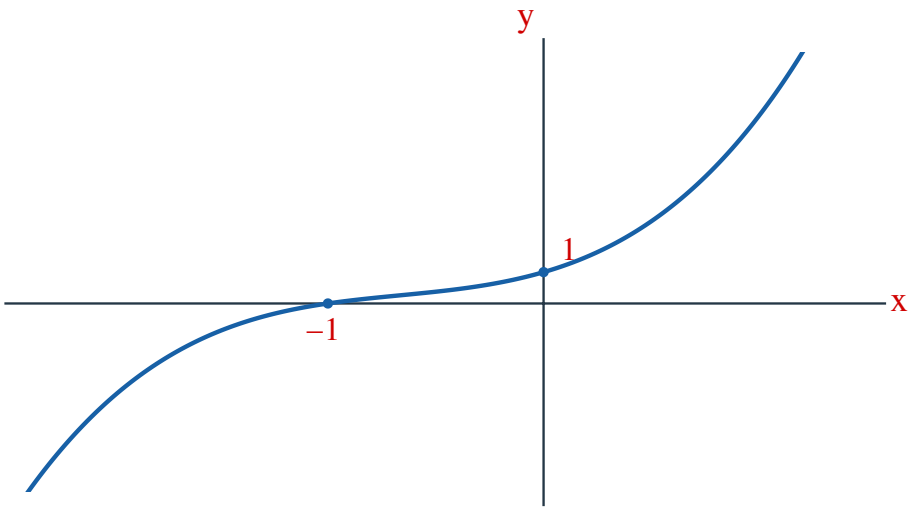
$y = (x+1)(x^2+x+1)$

Sketch features

The quadratic factor has discriminant $1 - 4 = -3$, so the only real root is -1.

y-intercept $(0, 1)$; left tail down, right tail up.

Sketch sketch



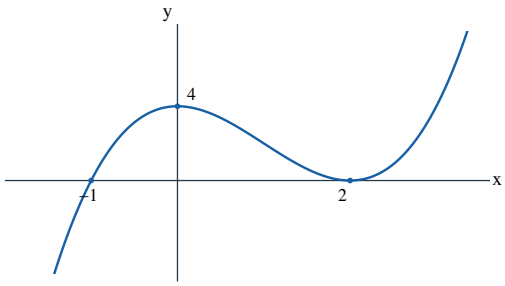
Later, differentiation confirms that there are no turning points:

$y\prime = 3x^2 + 4x + 2 = 3(x + \frac{2}{3})^2 + \frac{2}{3} > 0.$

Finding the equation yourself

Solution 10

A monic cubic crosses at $(-1, 0)$, touches at $(2, 0)$, and has a maximum at $(0, 4)$. Find a, b and c in $y = x^3 + ax^2 + bx + c$.



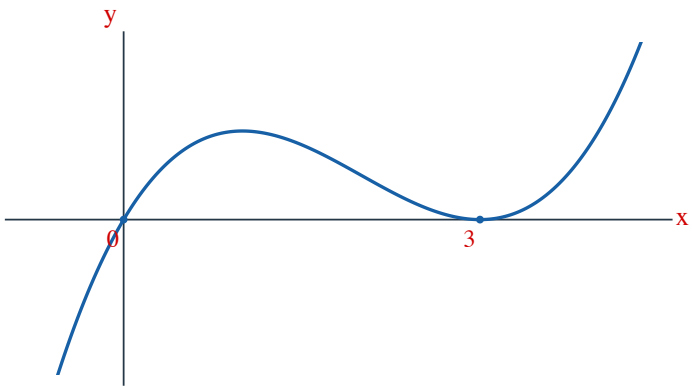
Answer

$$y = (x-2)^2(x+1)$$
$$= (x^2 - 4x + 4)(x+1)$$
$$= x^3 - 3x^2 + 4$$
$$a = -3, \quad b = 0, \quad c = 4$$

Test Your Understanding — Solution 11

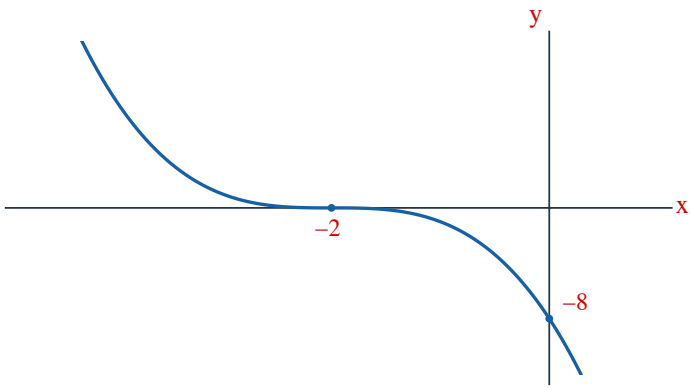
1. Sketch $y = x(x-3)^2$.

1



2. Sketch $y = -(x+2)^3$.

2



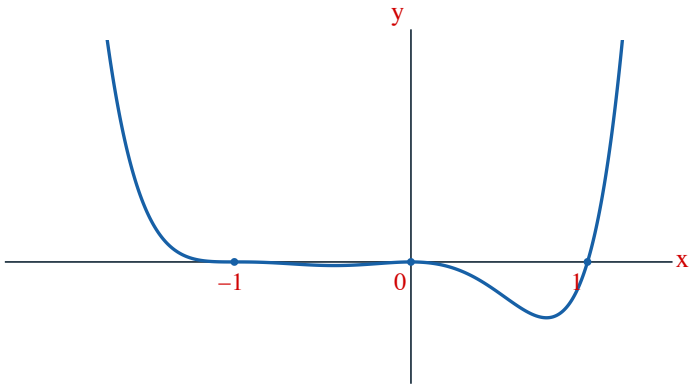
3. A positive cubic crosses at -2 and touches at 3. Give one suitable equation.

3

$y = (x-3)^2(x+2)$

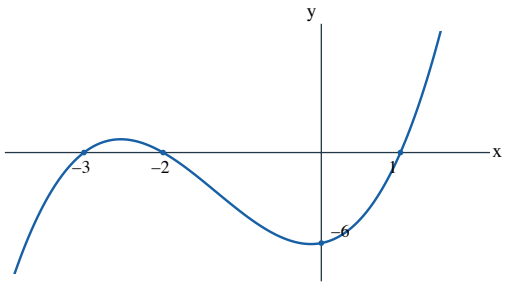
Extension. Sketch $y = 2x^2(x-1)(x+1)^3$.

Extension



Exercise 4A –Y1 Pure pages 63 — Solution 12

5. A monic cubic crosses the x-axis at -3, -2 and 1. (a) Find b, c and d in $y = x^3 + bx^2 + cx + d$. (b) Find its y-intercept.

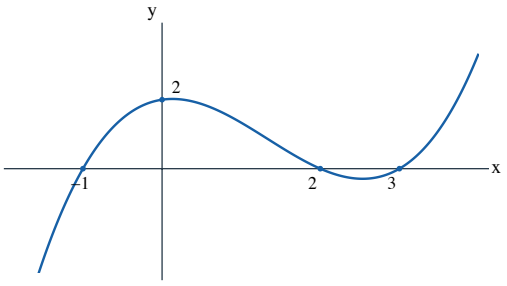


5

$y = (x+3)(x+2)(x-1) = x^3 + 4x^2 + x - 6$

$b = 4,\ c = 1,\ d = -6; \quad (0, -6).$

6. A cubic crosses at -1, 2 and 3 and has y-intercept 2. Find a, b, c, d in $y = ax^3 + bx^2 + cx + d$.



6

$2 = 6a \Rightarrow a = \frac{1}{3}$

$y = \frac{1}{3}(x+1)(x-2)(x-3)$

$a = \frac{1}{3},\ b = -\frac{4}{3},\ c = \frac{1}{3},\ d = 2$

Exercise 4A –Y1 Pure pages 63 — Solution 13

$f(x) = (x-10)(x^2-2x)+12x$

7(a) Express f(x) in the form $x(ax^2+bx+c)$.

7(a)

$f(x) = x^3 - 12x^2 + 32x = x(x^2-12x+32)$

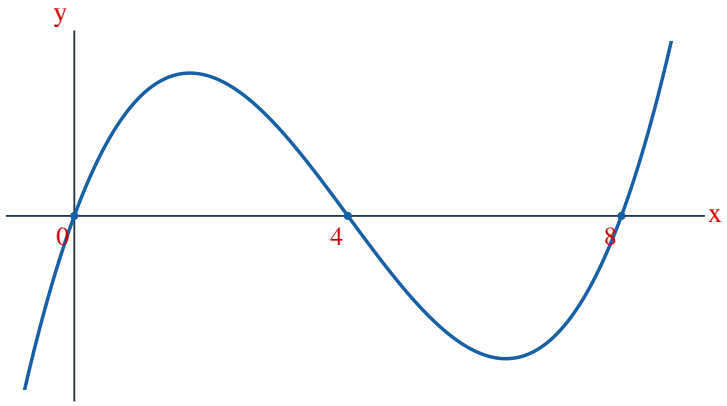
7(b) Factorise completely.

7(b)

$f(x) = x(x-4)(x-8)$

7(c). Sketch the curve, marking all intercepts.

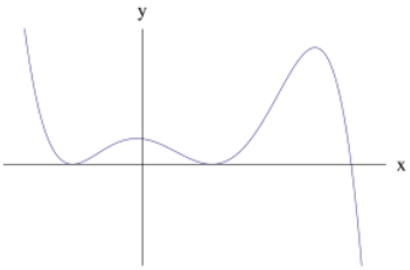
7(c)



Exercise 4A Extension — Solution 14

1. MAT 2012 1E

Which equation could give this sketch?



A : $(3-x)^2(3+x)^2(1-x)$

B : $-x^2(x-9)(x^2-3)$

C : $(x-6)(x-2)^2(x+2)^2$

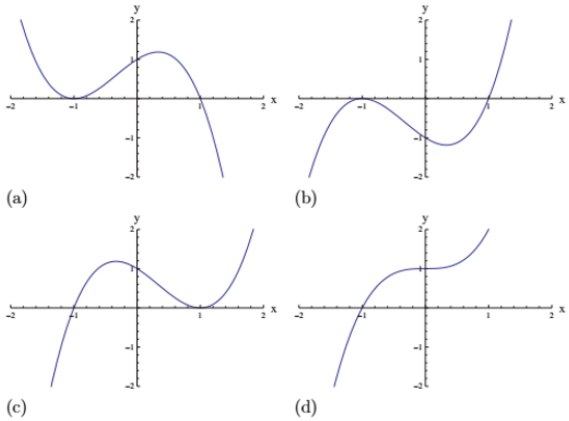
D : $(x^2-1)^2(3-x)$

1

D: two touching roots and a crossing root further right, with a negative leading coefficient.

2. MAT 2011 1A

Which sketch represents $y = x^3 - x^2 - x + 1$?



2

$x^2(x-1)-(x-1) = (x+1)(x-1)^2$

Crosses at -1, touches at 1; y-intercept 1. Answer (c).

Recap

Solution 15

For distinct a, b, c, consider $y = (x-a)(x-b)^2(x-c)^3$. What happens at each root?

a A simple root.

a

The graph crosses the x-axis at $x = a$.

b A double root.

b

The graph touches the x-axis at $x = b$.

c A triple root.

c

The graph crosses at $x = c$, with a horizontal tangent (stationary inflection).

Quartics

Solution 16

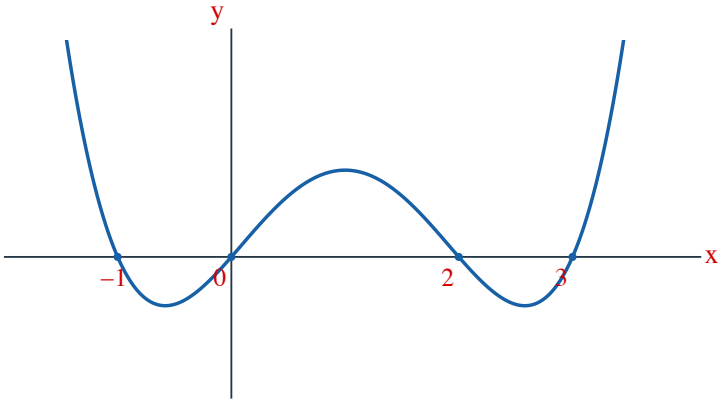
1 Sketch

$y = x(x+1)(x-2)(x-3)$

1 features

Both tails up; crosses at -1, 0, 2, 3. The y-intercept is 0.

1 sketch



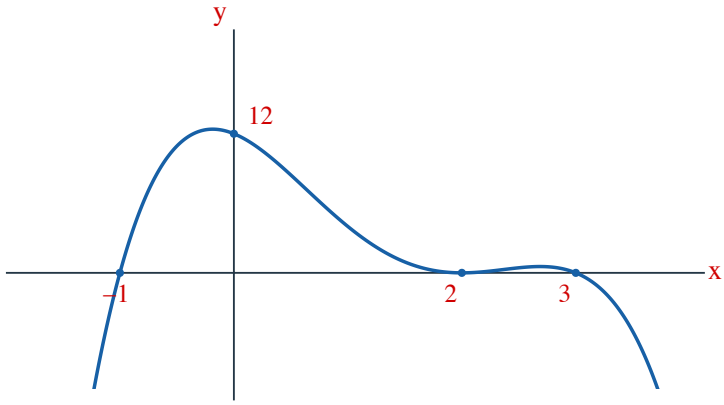
2 Sketch

$y = (x-2)^2(x+1)(3-x)$

2 features

Both tails down; crosses at -1 and 3, touches at 2. The y-intercept is 12.

2 sketch



Quartics

Solution 17

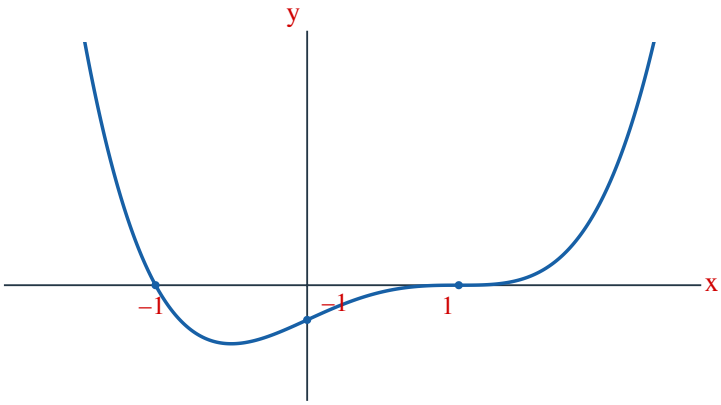
1 Sketch

$y = (x+1)(x-1)^3$

1 features

Both tails up. Crosses at -1 and at the triple root 1. y-intercept -1.

1 sketch



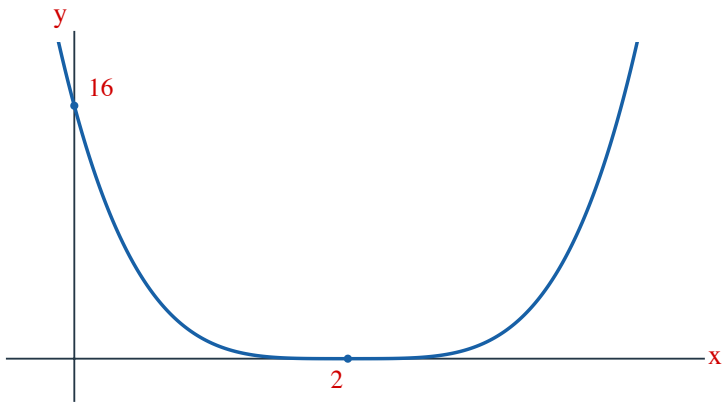
2 Sketch

$y = (x-2)^4$

2 features

Both tails up. Touches at the quadruple root 2; y-intercept 16.

2 sketch



Test Your Understanding

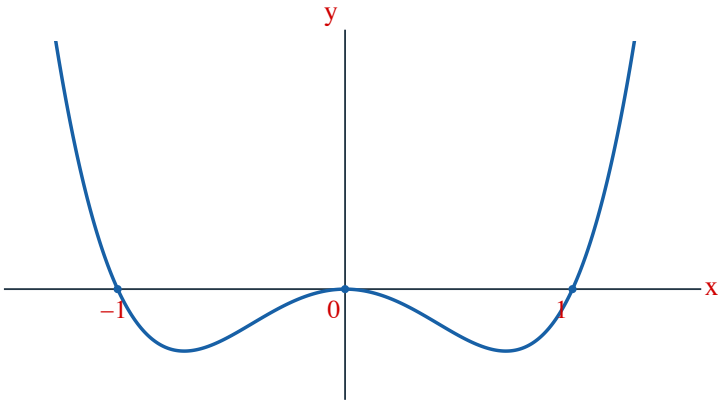
Solution 18

1 Sketch

$y = x^2(x+1)(x-1)$

1 features Both tails up; crosses at ± 1, touches at 0.

1 sketch

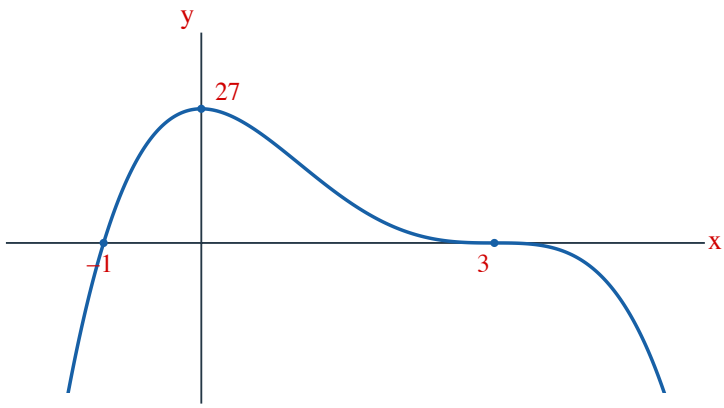


2 Sketch

$y = -(x+1)(x-3)^3$

2 features Both tails down; crosses at -1 and the triple root 3; y-intercept 27.

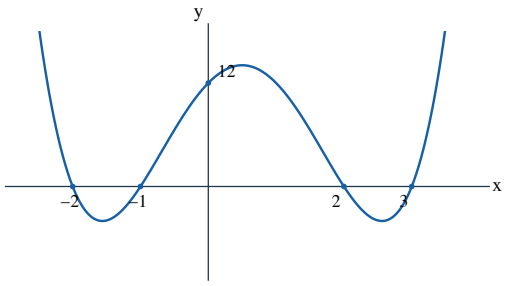
2 sketch



Exercise 4B – Pure Y1 Pages 65-66

Solution 19

3. The monic quartic $y = x^4 + bx^3 + cx^2 + dx + e$ has roots $-2, -1, 2, 3$. Find its y-intercept P and b, c, d, e.



3

$y = (x+2)(x+1)(x-2)(x-3)$

$y = x^4 - 2x^3 - 7x^2 + 8x + 12$

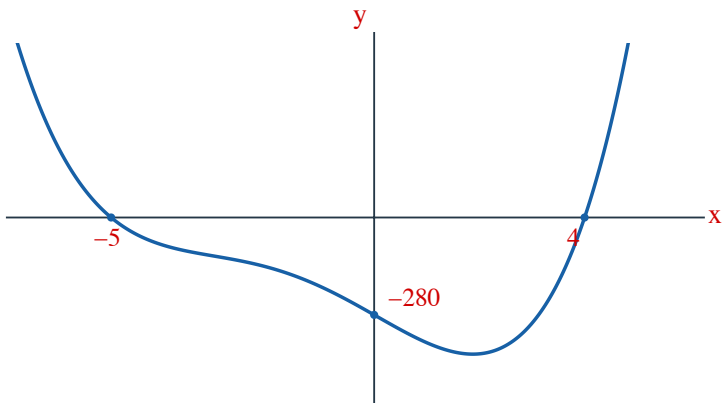
$P = (0, 12),\ b = -2,\ c = -7,\ d = 8,\ e = 12$

4 Sketch

$y = (x+5)(x-4)(x^2+5x+14)$

4 features The quadratic has discriminant -31. The only roots are -5 and 4; y-intercept -280. Both tails up.

4 sketch



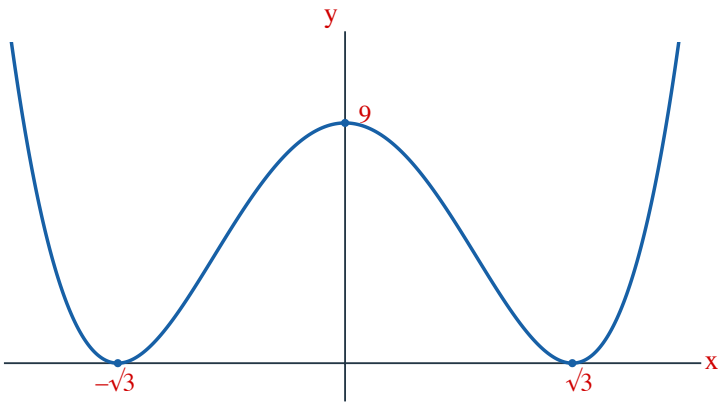
Exercise 4B Extension

Solution 20

STEP I 2012 Q2

(a) Sketch $y = x^4 - 6x^2 + 9$.

a



a features

$y = (x^2 - 3)^2 \geq 0$

Touches at $x = \pm\sqrt{3}$; y-intercept 9.

(b) For which b does $x^4 - 6x^2 + b = 0$ have each number of distinct real roots?

b

0 roots: $b > 9$.

1 root: never.

2 roots: $b = 9$ or $b < 0$.

3 roots: $b = 0$.

4 roots: $0 < b < 9$.

GCSE RECAP: Reciprocal Graphs

Solution 21

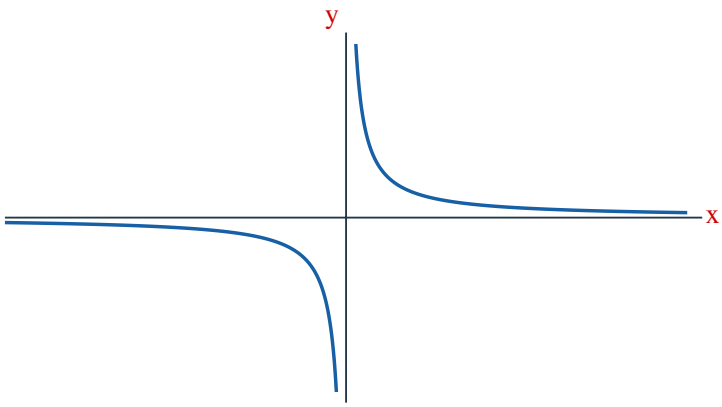
1 Sketch

$$y = \frac{1}{x}$$

1 features

Quadrants I and III; undefined at x=0.

1 sketch



y = 1/x

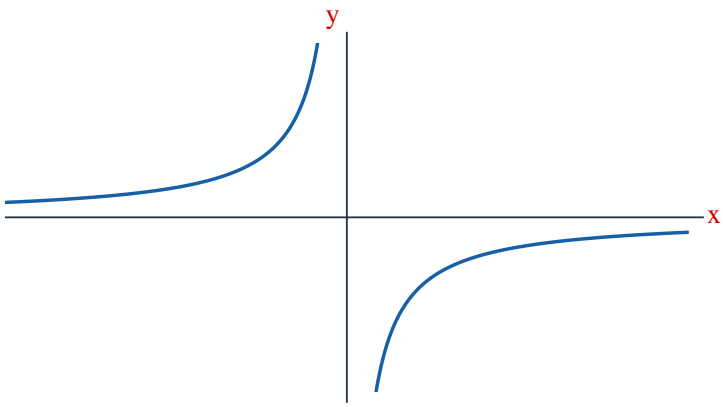
2 Sketch

$$y = -\frac{3}{x}$$

2 features

Quadrants II and IV; undefined at x=0.

2 sketch



y = -3/x

Both graphs have asymptotes $x = 0$ and $y = 0$. An asymptote is a line approached by a curve. These reciprocal graphs do not meet their asymptotes.

Reciprocal Graphs

Solution 22

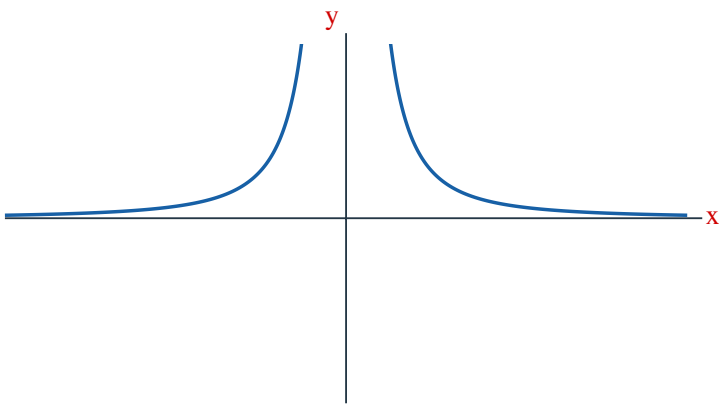
1 Sketch

$$y = \frac{3}{x^2}$$

1 features

Positive on both sides; symmetric in the y-axis.

1 sketch



y = 3/x²

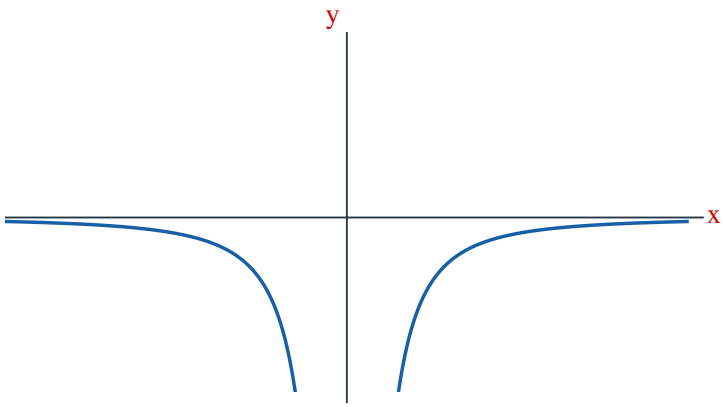
2 Sketch

$$y = -\frac{4}{x^2}$$

2 features

Negative on both sides; symmetric in the y-axis.

2 sketch



y = -4/x²

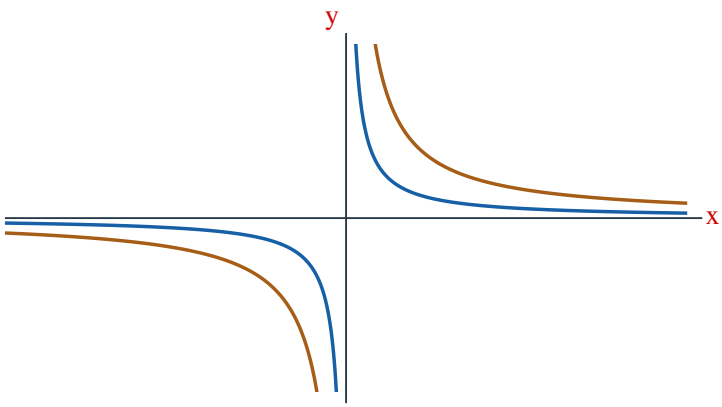
For both graphs the asymptotes are $x = 0$ and $y = 0$.

Reciprocal Graphs

Solution 23

On the same axes sketch $y = 1/x$ and $y = 3/x$.

Sketch



y = 1/x y = 3/x

Compare

For the same nonzero x, the ordinate of $3/x$ is three times that of $1/x$.

For positive x it lies higher; for negative x it lies lower.

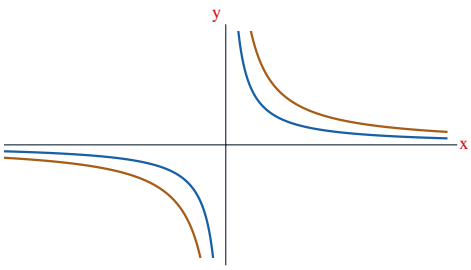
Exercise 4C – Pure Y1 page 67

Solution 24

1. Sketch each pair on a separate set of axes. Blue is the first curve; brown is the second.

$y = 2/x, \quad y = 4/x$

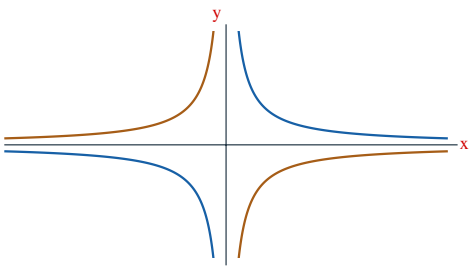
a



y = 2/x y = 4/x

$y = 2/x, \quad y = -2/x$

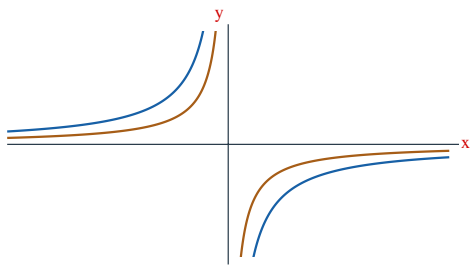
b



y = 2/x y = -2/x

$y = -4/x, \quad y = -2/x$

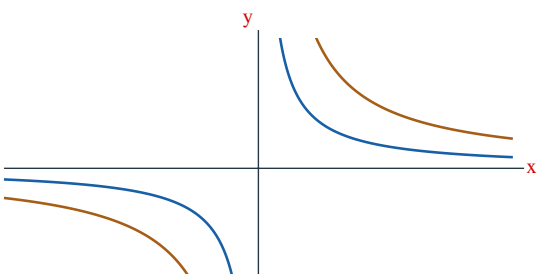
c



y = -4/x y = -2/x

$y = 3/x, \quad y = 8/x$

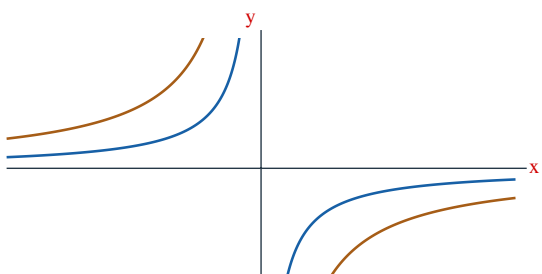
d



y = 3/x y = 8/x

$y = -3/x, \quad y = -8/x$

e



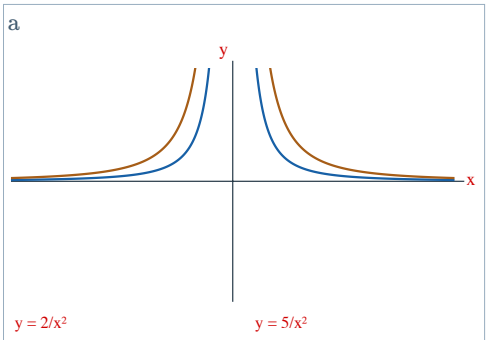
y = -3/x y = -8/x

Exercise 4C – Pure Y1 page 67 — Solution 25

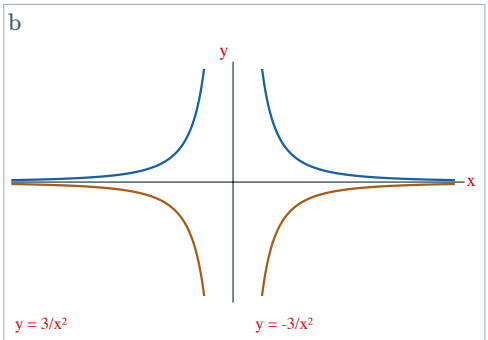
2. Sketch each pair. Blue is the first curve; brown is the second. All have asymptotes x=0 and y=0.

$y = 2/x^2, \quad y = 5/x^2$ $\qquad$ $y = 3/x^2, \quad y = -3/x^2$ $\qquad$ $y = -2/x^2, \quad y = -6/x^2$

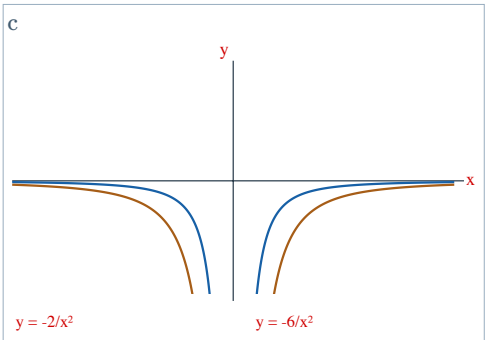
a



b

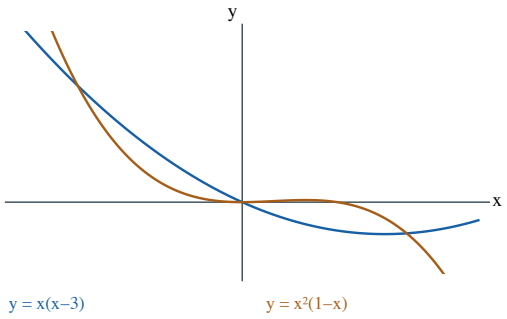


c



Points of Intersection — Solution 26

Find the intersections of $y = x(x-3)$ and $y = x^2(1-x)$.



Intersections

$$x^2 - 3x = x^2 - x^3$$

$$x(x^2 - 3) = 0$$

$$x = 0, \pm\sqrt{3}$$

$$(0,0),\ (\sqrt{3}, 3 - 3\sqrt{3}),\ (-\sqrt{3}, 3 + 3\sqrt{3})$$

Do not divide by x: that would lose the solution x=0.

Further example involving unknown constants

Solution 27

Sketch $y = x^2(3x - a)$ and $y = b/x$, where a and b are positive. How many intersections are there?

Sketch

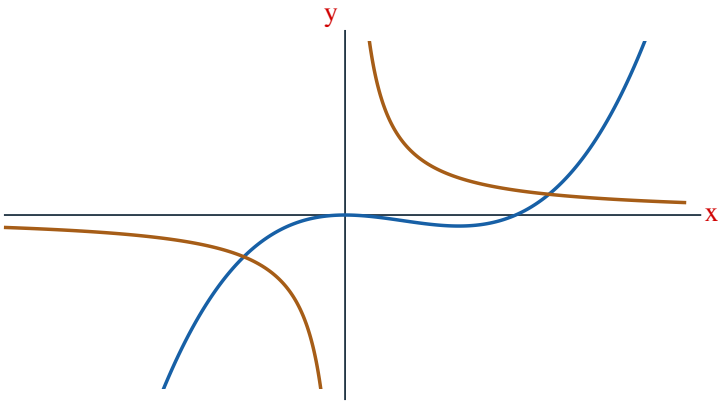


Illustration: a = 3 b = 1

Count

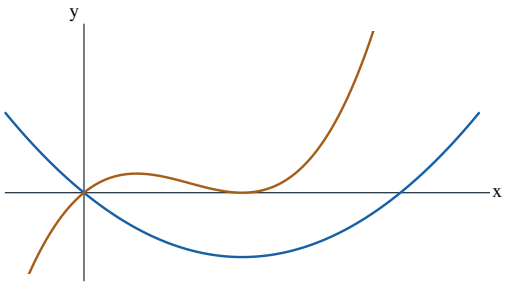
The cubic touches at 0 and crosses at a/3. The reciprocal has branches in quadrants I and III.

There is one intersection for x<0 and one for x>a/3: **two real intersections**.

Test Your Understanding

Solution 28

Sketch $y = x(x - 4)$ and $y = x(x - 2)^2$. Find all intersections.



Answer

$$x(x - 4) = x(x - 2)^2$$

$$x(x^2 - 5x + 8) = 0$$

The quadratic factor has discriminant $25 - 32 = -7$.

The only intersection is $(0, 0)$.

Exercise 4D – Pure 1 Page 69-71 Solution 29

Sketch $y = x^2(x + a)$ and $y = b/x$, where a,b>0. State the number of intersections and form an equation for their x-coordinates.

Sketch

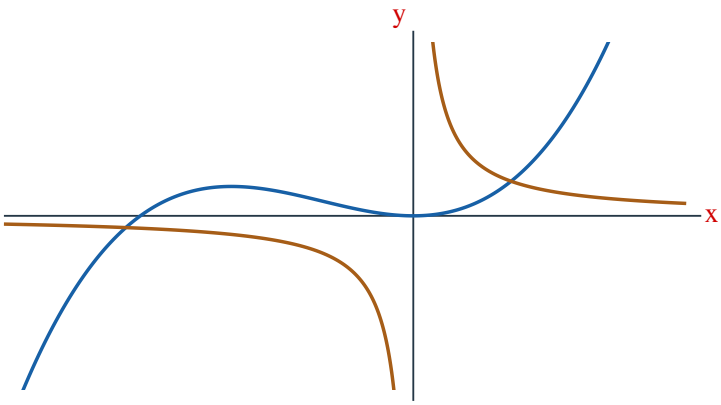


Illustration: a = 2 b = 1

Answer

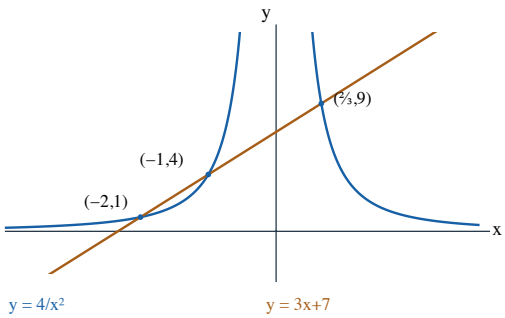
The cubic crosses at −a and touches at 0. There is one intersection for x<−a and one for x>0.

$$x^2(x + a) = \frac{b}{x} \Rightarrow x^4 + ax^3 - b = 0$$

There are two real intersections.

Exercise 4D – Pure 1 Page 69-71 Solution 30

Find the intersections of $y = 4/x^2$ and $y = 3x + 7$.



Answer

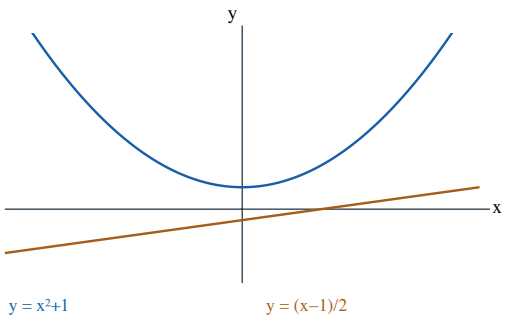
$$4 = 3x^3 + 7x^2$$

$$0 = (x + 1)(x + 2)(3x - 2)$$

$$(-2, 1), \quad (-1, 4), \quad (\tfrac{2}{3}, 9)$$

Exercise 4D – Pure 1 Page 69-71 Solution 31

1(a) Show that $y = x^2 + 1$ and $y = (x - 1)/2$ never meet.



a

$2x^2 - x + 3 = 0, \quad \Delta = 1 - 24 = -23 < 0$

1(b) For which a do $y = x^2 + a$ and $y = (x - 1)/2$ meet at two points?

1(b)

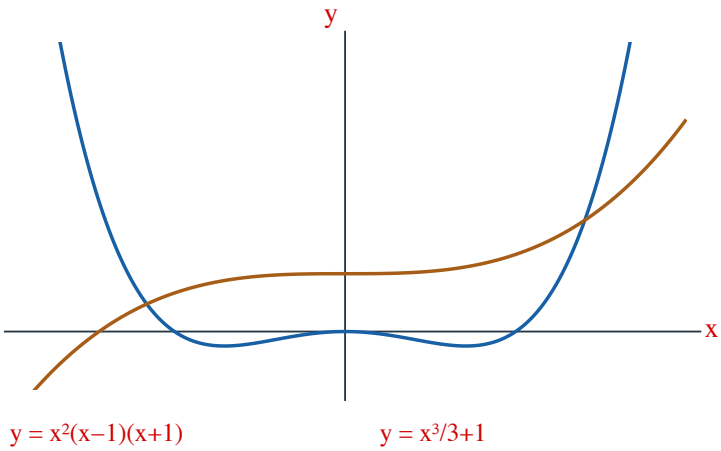
$2x^2 - x + 2a + 1 = 0$

$1 - 8(2a + 1) > 0 \Rightarrow a < -\frac{7}{16}$

Exercise 4D – Pure 1 Page 69-71 Solution 32

Sketch $y = x^2(x - 1)(x + 1)$ and $y = x^3/3 + 1$ on the same axes. How many real solutions does $x^2(x - 1)(x + 1) = x^3/3 + 1$ have?

Sketch



Answer

The sketches intersect twice, once at a negative x and once at a positive x. There are **two real solutions**.

Exercise 4D - Extension

Solution 33

1. MAT 2005 1B

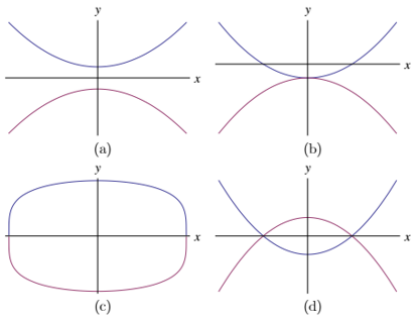
$(x^2+1)^{10} = 2x - x^2 - 2$

Does it have (A) solution 2, (B) no real solutions, (C) an odd number of real solutions, or (D) twenty real solutions?

1. MAT 2005 1B

The left side is at least 1. The right side is $-(x-1)^2 - 1 \leq -1$. Thus B: no real solutions.

2. MAT 2013 1D Which sketch represents $x^4 - y^2 = 2y + 1$?



2. MAT 2013 1D

$x^4 = (y+1)^2 \Rightarrow y = x^2 - 1$ or $y = -x^2 - 1$

Answer (b).

3. MAT 2010 1A For which k does $y = kx$ intersect $y = (x-1)^2$?

3. MAT 2010 1A

$x^2 - (k+2)x + 1 = 0$

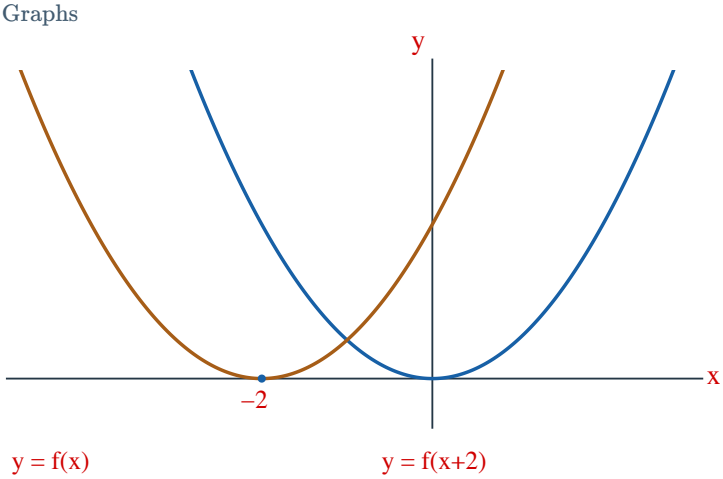
$\Delta = k(k+4) \geq 0 \Rightarrow k \leq -4$ or $k \geq 0$

Transformations of Functions

Solution 34

For $f(x) = x^2$, compare $y = f(x)$ with $y = f(x+2)$.

Graphs



Transformation

$f(x+2) = (x+2)^2$

Translation by $\binom{-2}{0}$: two units left. The vertex moves from $(0,0)$ to $(-2,0)$.

Transformations of Functions

Solution 35

Changes **inside** f affect x-coordinates inversely. Changes **outside** f act directly on y-coordinates.

1

$y = f(x - 3)$

1

Translate 3 units right: $(x, y) \mapsto (x + 3, y)$.

2

$y = f(x) + 4$

2

Translate 4 units up: $(x, y) \mapsto (x, y + 4)$.

3

$y = f(5x)$

3

Horizontal scale factor 1/5: $(x, y) \mapsto (x/5, y)$.

4

$y = 2f(x)$

4

Vertical scale factor 2: $(x, y) \mapsto (x, 2y)$.

Sketching transformed graphs

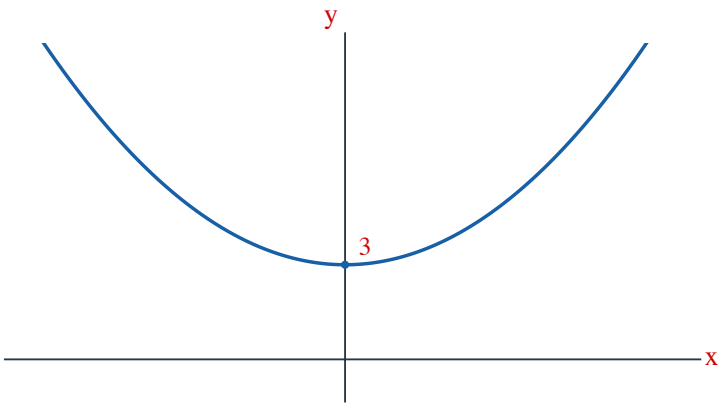
Solution 36

1 Sketch

$y = x^2 + 3$

1 features Translate $y = x^2$ up 3. Minimum and y-intercept (0,3); no x-intercept.

1 sketch

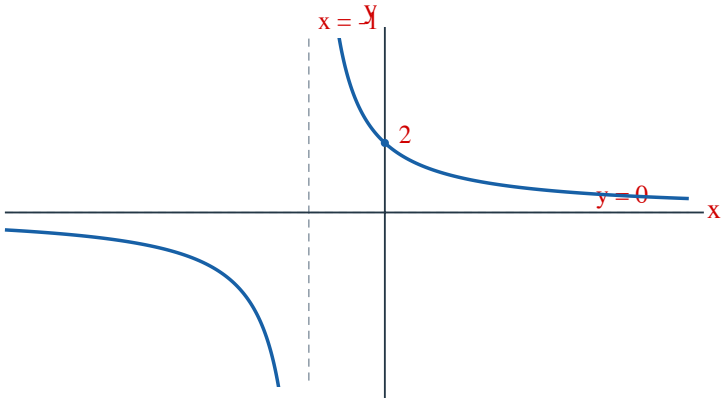


2 Sketch

$y = \dfrac{2}{x + 1}$

2 features Translate $y = 2/x$ left 1. Asymptotes x=−1, y=0; y-intercept (0,2); no x-intercept.

2 sketch



y = 2/(x+1)

More Examples

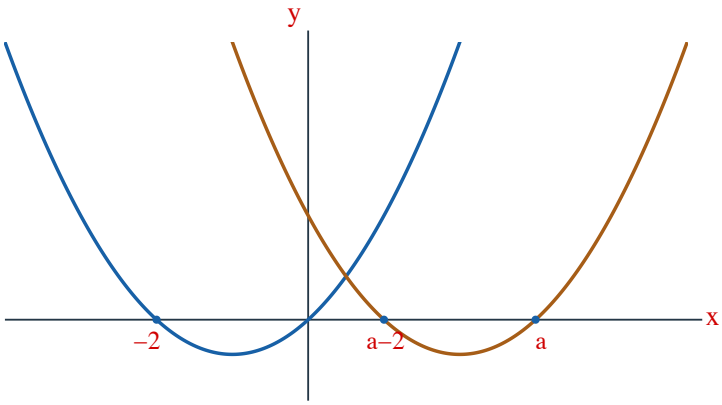
Solution 37

1. Sketch $f(x) = x(x+2)$ and $y = f(x-a)$, a>2.

1

Translate right a. New roots a−2 and a; y-intercept $a(a-2)$.

1 sketch



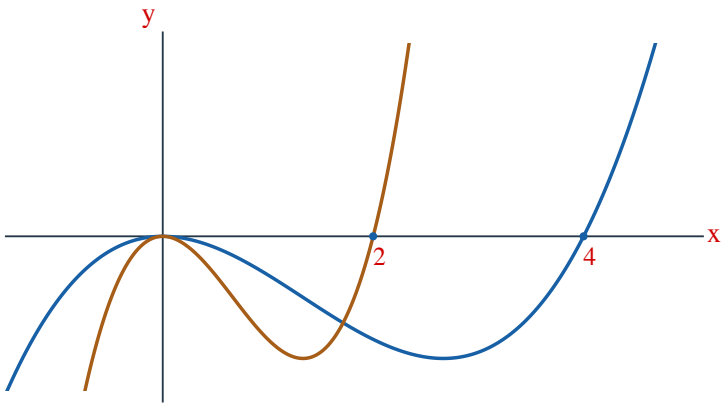
y = f(x) Illustration: a = 3

2. Sketch $f(x) = x^2(x-4)$ and $y = f(2x)$.

2

Halve all x-coordinates. The double root stays at 0; the root 4 moves to 2.

2 sketch



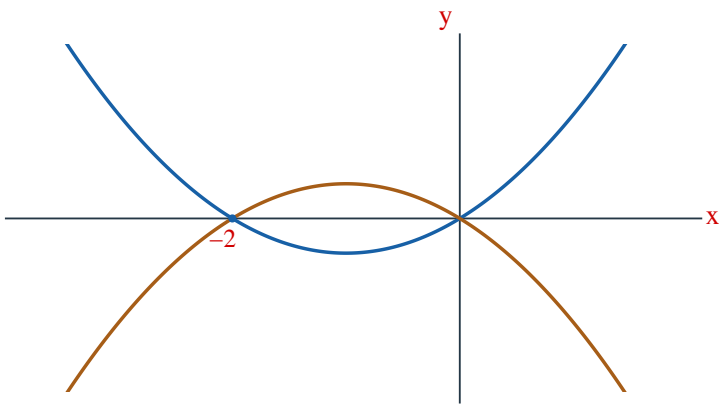
y = f(x) y = f(2x)

Reflections of Graphs

Solution 38

If $f(x) = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$.

Sketch



y = f(x) y = –f(x)

Answer

The negative sign is outside f: negate every y-coordinate. This is reflection in the x-axis.

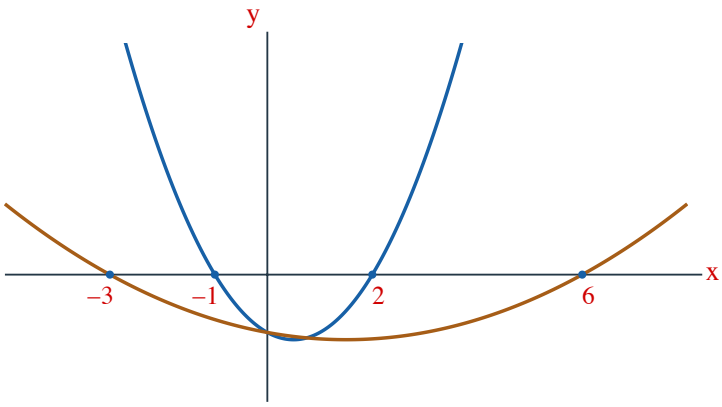
The roots −2 and 0 do not move.

Test Your Understanding — Solution 39

1. Sketch $f(x) = (x+1)(x-2)$ and $y = f(x/3)$.

1 Horizontal scale factor 3. Roots -3 and 6; y-intercept remains -2.

1 sketch



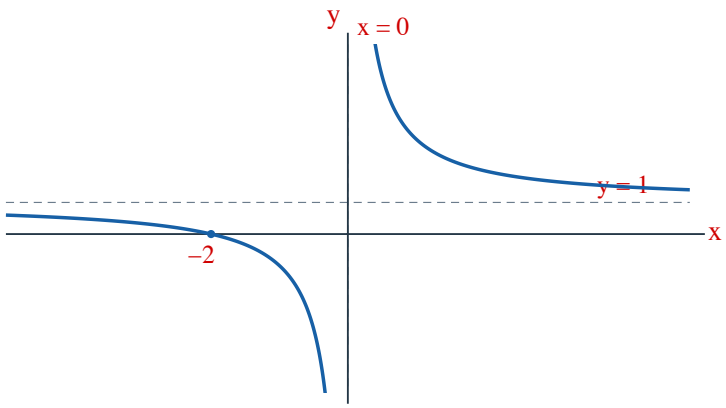
y = f(x) y = f(x/3)

2 Sketch

$$y = \frac{2}{x} + 1$$

2 features Translate $y = 2/x$ up 1. Asymptotes x=0 and y=1. Root -2; no y-intercept.

2 sketch



y = 2/x+1

Exercise 4E – Pure Y1 page 74-75 (translations) — Solution 40

7. P(4,−1) lies on $y = f(x)$. Find its image.

7(a)

$y = f(x-2)$

7(a)

(6, −1)

Two units right.

7(b)

$y = f(x) + 3$

7(b)

(4, 2)

Three units up.

8 Translate $y = 1/x$ so its asymptotes are x=4 and y=0. Give its new equation.

8

$$y = \frac{1}{x-4}$$

Four units right.

Exercise 4E – Pure Y1 page 74-75 (translations) Solution 41

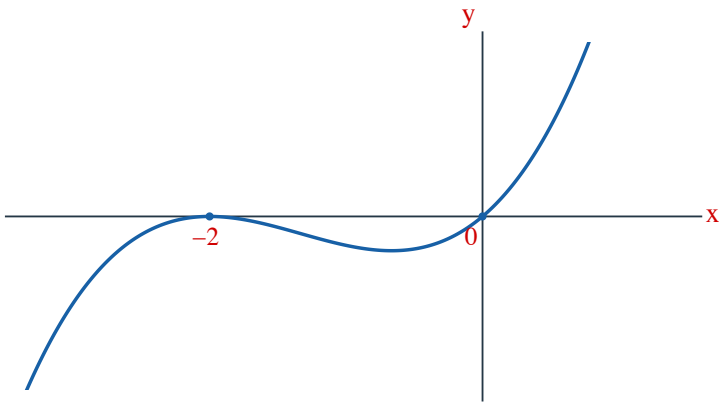
11(a) Sketch $y = x^3 + 4x^2 + 4x$.

a

$y = x(x+2)^2$

Touches at -2; crosses at 0.

a sketch



11(b) Find both a for which $(-1,0)$ lies on $y = (x+a)^3 + 4(x+a)^2 + 4(x+a)$.

11(b)

At x=−1, the input must be a root of f:

$-1 + a = 0$ or $-1 + a = -2$

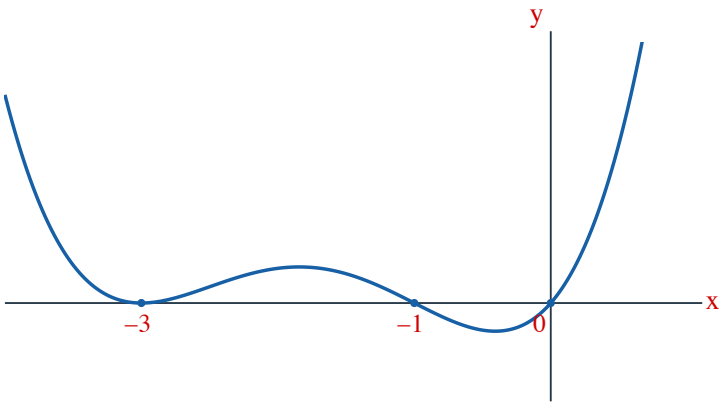
$a = 1$ or $a = -1$

Exercise 4E – Pure Y1 page 74-75 (translations) Solution 42

12(a) Sketch $y = x(x+1)(x+3)^2$.

a Touches at -3; crosses at -1 and 0. Both tails up.

a sketch



12(b) Find all b for which $(2,0)$ lies on $y = (x+b)(x+b+1)(x+b+3)^2$.

12(b)

$2 + b = 0, -1, -3$

$b = -2, -3, -5$

Exercise 4F – Pure 1 Page 78 (stretches/reflections) Solution 43

7. P(2,−3) lies on $y = f(x)$. Find its image.

7(a)

$y = f(2x)$

7(a)

$(1, -3)$

Halve the x-coordinate.

7(b)

$y = 4f(x)$

7(b)

$(2, -12)$

Multiply the y-coordinate by 4.

8 Q(−2,8) lies on $y = f(x)$. Find its image on $y = f(x/2)$.

8

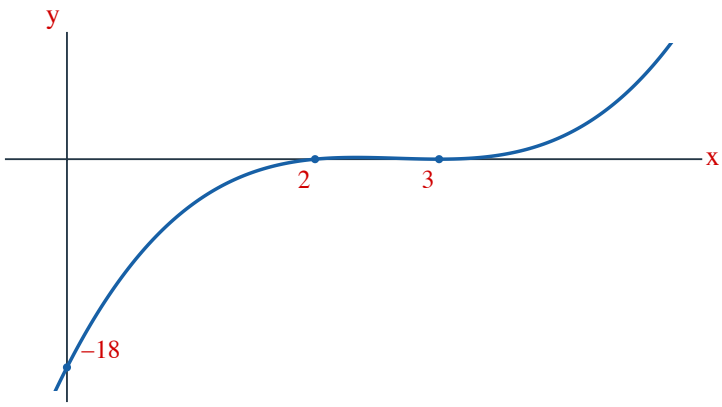
$(-4, 8)$

Double the x-coordinate.

Exercise 4F – Pure 1 Page 78 (stretches/reflections) Solution 44

9(a) Sketch $y = (x-2)(x-3)^2$.

a



9(b) The graph $y = (ax-2)(ax-3)^2$ passes through (1,0). Find both a.

9(b)

$(a-2)(a-3)^2 = 0 \Rightarrow a = 2$ or $a = 3$

Effect of transformation on specific points

Solution 45

Track these three points, in order: $(4,3),\ (1,0),\ (6,-4)$.

1

$y = f(x+1)$

1

$(3,3),\ (0,0),\ (5,-4)$

2

$y = f(2x)$

2

$(2,3),\ (\frac{1}{2},0),\ (3,-4)$

3

$y = 3f(x)$

3

$(4,9),\ (1,0),\ (6,-12)$

4

$y = f(x) - 1$

4

$(4,2),\ (1,-1),\ (6,-5)$

5

$y = f(x/4)$

5

$(16,3),\ (4,0),\ (24,-4)$

6

$y = f(-x)$

6

$(-4,3),\ (-1,0),\ (-6,-4)$

7

$y = -f(x)$

7

$(4,-3),\ (1,0),\ (6,4)$

Test Your Understanding

Solution 46

For $f(x) = x^2(9-2x)$, the minimum is (0,0) and maximum is (3,27).

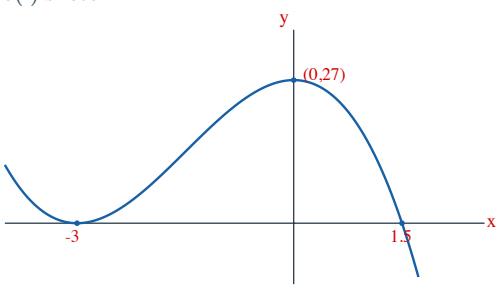
a Find the other x-intercept A.

a $x^2(9-2x) = 0 \Rightarrow A = (\frac{9}{2},0)$

b(i) Sketch $y = f(x+3)$. Mark extrema and intercepts.

b(i) Minimum $(-3,0)$, maximum and y-intercept (0,27), other root (3/2,0).

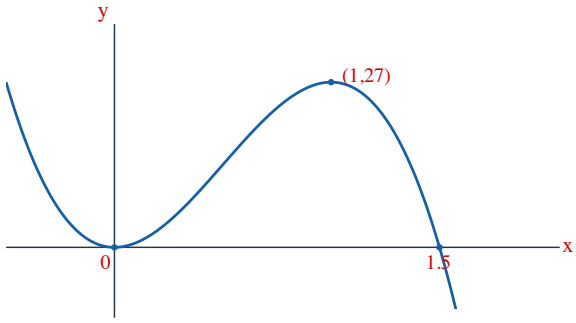
b(i) sketch



b(ii) Sketch $y = f(3x)$. Mark extrema and intercepts.

b(ii) Minimum and y-intercept (0,0), maximum (1,27), other root (3/2,0).

b(ii) sketch



c $y = f(x) + k$ has maximum (3,10). Find k.

c $27 + k = 10 \Rightarrow k = -17$

Exercise 4G Pure Y1 Pages 80 — Solution 47

5. P(2,1) lies on $y = f(x)$.

a On $y = f(ax)$, P maps to Q(4,1). Find a.

a

$2/a = 4 \Rightarrow a = \frac{1}{2}$

b(i) Find the image on $y = f(x - 4)$.

b(i)

$(6, 1)$

b(ii) Find the image on $y = 3f(x)$.

b(ii)

$(2, 3)$

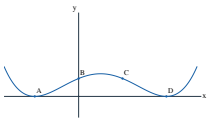
b(iii) Find the image on $y = f(x)/2 - 4$.

b(iii)

$(2, -\frac{7}{2})$

Exercise 4G Pure Y1 Pages 80 — Solution 48

6. A curve $y = f(x)$ has points A(−1,0), B(0,2), C(1,2), D(2,0). Sketch each transformation and give the images of A, B, C, D.

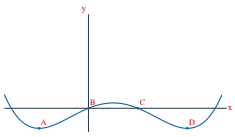


$y + 2 = f(x)$

a

$A(-1, -2),\ B(0, 0)$
$C(1, 0),\ D(2, -2)$

a sketch

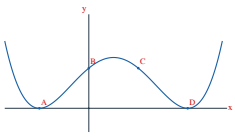


$\frac{1}{2}y = f(x)$

b

$A(-1, 0),\ B(0, 4)$
$C(1, 4),\ D(2, 0)$

b sketch

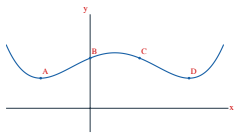


$y - 3 = f(x)$

c

$A(-1, 3),\ B(0, 5)$
$C(1, 5),\ D(2, 3)$

c sketch

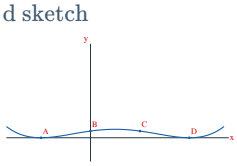


$3y = f(x)$

d

$A(-1, 0),\ B(0, \frac{2}{3})$
$C(1, \frac{2}{3}),\ D(2, 0)$

d sketch



$2y - 1 = f(x)$

e

$A(-1, \frac{1}{2}),\ B(0, \frac{3}{2})$
$C(1, \frac{3}{2}),\ D(2, \frac{1}{2})$

e sketch

